Chinmay Modi

chinmaymodi@gmail.com | github.com/chinmaymodi | +91 93168 89468
chinmaymodi.github.io | linkedin.com/in/chinmaymodi

PROFESSIONAL SUMMARY

Product Engineer with 4+ years of experience building enterprise applications and distributed systems using C#, .NET, Python, React, PostgreSQL, and Redis. Experienced in modernizing legacy platforms, developing scalable APIs and microservices, optimizing database performance, and building production-grade applications with automated testing and CI/CD. Recently completed an M.S. in Data Analytics with continued hands-on development through full-stack and machine learning projects. Directs AI-assisted engineering workflows with human ownership of architecture, data integrity, and acceptance decisions using coding agents for generation while owning every design and correctness call.

TECHNICAL SKILLS

- Languages: C#, Python, JavaScript, SQL
- Frameworks: .NET, React, MVC, Entity Framework
- Web & APIs: REST APIs, Swagger, Postman
- Databases: PostgreSQL, MSSQL, MongoDB, Redis
- DevOps & Tools: Git, Jenkins, SonarQube, WhiteSource, SumoLogic, Docker, Kubernetes, CI/CD, Testcontainers, xUnit, Moq
- AI-Assisted Engineering: LLM integration, coding agents, automated code review, test generation, debugging workflows, AI-directed engineering workflow

PROFESSIONAL EXPERIENCE

Product Engineer – Independent Project (NSE Strategy Platform), Feb 2026 – Present

- Designed and built a full-stack algorithmic trading platform for family investment operations using ASP.NET Core 10, React 19, TypeScript, PostgreSQL, and Redis, including REST APIs, scheduled data ingestion, caching, and a production-oriented Windows service deployment.
- Engineered ML signal pipelines using 27+ cross-sectional features, walk-forward validation, ensemble voting, and out-of-sample Sharpe-based model evaluation; processed and backfilled 5.4M+ options data records.
- Built automated backend and frontend testing using xUnit, Moq, and Testcontainers, with 900+ automated tests across C# and Python covering application, integration, and strategy workflows; C# line coverage 96.3%, Python library coverage 92.6%.
- Implemented safeguards preventing accidental live broker order placement during builds and tests, reducing operational risk during development.
- Built end-to-end with an AI-assisted engineering workflow, directing multi-agent planning, code-review, debugging, and domain-research agents while owning all architecture, data-validation, and acceptance decisions.
- Deployed the platform as a persistent Windows service with scheduled daily data collection, missed-slot catch-up, and automatic restart on failure; built an options execution engine with realistic fill models and paper-trading evaluation.
- Performed all data engineering and validation: adapted to the NSE bhavcopy format change, handled corporate-action re-basing, and wrote verification scripts that audited 5.4M+ options records for gaps and duplicates, ensuring AI-assisted analysis ran on validated data.
- Defined product scope and prioritized features across ingestion, backtesting, and execution; rejected speculative strategies based on out-of-sample validation results.

Software Developer II — SMS Assist LLC, Chicago, Feb 2022 – Nov 2022

- Contributed to the modernization of a legacy monolithic application by designing and implementing C#/.NET microservices, refactoring tightly coupled modules into independently deployable services. This improved system modularity and enabled more stable release cycles.
- Resolved high-volume transactional query latency by refactoring inefficient LINQ logic, optimizing ORM mappings, introducing targeted PostgreSQL indexing strategies, and leveraging prepared statements. These changes significantly improved production data access performance without altering business logic.
- Developed solutions for caching results on Redis, to prevent repeated database lookups. This assisted in developing file upload and processing functionality for clients improving throughput and reducing additional costly client attempts made in error.
- Deployed updated modules using Jenkins. Strengthened engineering quality by expanding unit and integration test coverage and embedding automated test validation into CI pipelines, increasing code coverage and reducing regression defects during deployments. Participated in code reviews and drove continuous improvement of development practices.
- Improved code health and security posture by addressing SonarQube findings, enforcing quality gates, and remediating vulnerable dependencies through WhiteSource, reducing technical debt and strengthening compliance standards.
- Supported production stability during on-call rotations by diagnosing incidents using SumoLogic log analysis and collaborating cross-functionally to restore services quickly, minimizing downtime impact.
- Wrote performant front-end code in React, as well as updating and maintaining legacy code in KnockoutJS.
- Created demos on short notice for client onboarding to show potential ideas for fulfilling their specific needs. Under this responsibility, I worked with a small team on implementing rapidly evolving client requirements into working MVP demos.

Software Developer — Ab Ovo North America, Boston, Apr 2019 – Jan 2022

- Played a key role in a multi-year enterprise rail cargo planning implementation for a major Canadian client, evolving from core developer to hybrid Technical Consultant and code owner. This transition helped increase US/Canada collaboration and improved responsiveness during later project phases.
- Developed EDI interfaces, modeled logistics network components, designed internal data retrieval APIs, and integrated third-party REST services enabling seamless cross-system data exchange; built a load and performance testing framework from scratch to identify memory bottlenecks.
- As Technical Consultant and code owner, resolved client-side performance issues, designed scalable interface enhancements, and implemented dataset partitioning and archiving strategies to handle growing data volumes.
- Mentored junior developers, authored technical documentation, and established debugging best practices, reducing ramp-up time and increasing team productivity.

EDUCATION

- Master of Science in Data Analytics — McDaniel College (2023–2025), GPA 4.0
  Capstone: Earthquake Damage Prediction - XGBoost with spatial feature engineering, SHAP interpretability (weighted F1 0.753)
- Master of Science in Computer Science — University of Houston–Clear Lake (Dec 2017), GPA 3.95
- Bachelor of Technology in Information and Communication Technology — DA-IICT, Gujarat, India (Jun 2015)

CERTIFICATIONS

- Certified Scrum Product Owner (CSPO)